Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
F		546567

TITLE

Summary

BHSBVHBHSBVSDBJVBJDSBJVBSDJBVJDBBVJBDJVBJDB

Keywords: SDCBJHDSBJCDSJBJCDBSJVBSDB;SDCHJSBVSD;SDCDSC;SCSDCDSCS

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1 Introduction

1.1 Background

good

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good

good

1.2 Literature Review

good

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good

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good

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good

1.3 The Description of the Problem

The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law. The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

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1.4 Our work

2 title

2.1 title

2.1.1 title

1.title

3 Assumptions

Assumption 1: The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

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- good
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      - good
      - good

1. good
2. good
   (a) good
   (b) good
      i. good
      ii. good
         A. good
         B. good

4 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: **Notations used in this literature**

Symbol	Description
PSNR	Peak signal-to-noise ratio for image quality evaluation
UCIQE	Underwater Color Image Quality Evaluation metric
UIQM	Underwater Image Quality Measure

Table 2: table

A	B	C
1	2	3
4	5	6

The core symbols and their definitions used in this study are summarized in Table2, providing an overview of the key parameters and their related meanings.

Table 3: table

Symbol	Description	C
PSNR	Peak signal-to-noise ratio for image quality evaluation	C



Figure 1: fig

4.1 Proof of Lagrange Mean Value Theorem

4.1.1 Step 1: Construct the Auxiliary Function

To apply Rolle's Theorem, we construct an auxiliary function $F(x)$, whose geometric meaning is "the difference between the ordinate of the curve $y = f(x)$ and the ordinate of the chord AB" (where $A(a, f(a))$ and $B(b, f(b))$ are the endpoints of the curve on $[a, b]$).

The equation of the straight line passing through A and B is:

$$y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$$

which can be rewritten as:

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

$$\int x^n \mathrm{dx} = \frac{\mathrm{x^{n+1}}}{\mathrm{n+1}} + \mathrm{C}$$

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

Thus, the auxiliary function is defined as:

$$F(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right]$$

To apply Rolle's Theorem, we construct an auxiliary function $F(x)$, whose geometric meaning is "the difference between

$$F(x)$$

$$F(x)$$

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \tag{1}$$

5 Models

5.1 Modeling and Solving of Problem 1

5.1.1 Problem Analysis

5.1.2 Model Preparation

5.1.3 Model Construction

5.1.4 Model Solution

5.2 Modeling and Solving of Problem 1

5.2.1 Problem Analysis

5.2.2 Model Preparation

5.2.3 Model Construction

5.2.4 Model Solution

5.3 Modeling and Solving of Problem 1

5.3.1 Problem Analysis

5.3.2 Model Preparation

5.3.3 Model Construction

5.3.4 Model Solution

6 Sensitivity Analysis

7 Model Evaluation and Promotion

7.1 Model Evaluation

7.1.1 Advantages

7.1.2 Limitations

7.2 Future Work

7.2.1 Model extension

7.2.2 Model application

8 Conclusions

[2]

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Appendices

Report on Use of AI

1. Bing AI
   Query1:
   Output:

2. Bing AI
   Query1:
   Output:

3. Bing AI
   Query1:
   Output:

4. Bing AI
   Query1:
   Output: